

AWK Print Header Dynamically

in a awk script if we don't know the number of fields then how to print the header ?

In an `awk` script, if you **don't know the number of fields in advance**, but still want to print the **header (field names)** from the **first line (usually)**, you can loop through the fields dynamically using `NF` (Number of Fields). Here's how you can do it:

Example

Suppose your input file is like this:

```
txt

Name Age Gender Country
Alice 24 F USA
Bob 30 M UK
```

AWK Script to Print the Header

```
bash

awk 'NR==1 {
    for (i = 1; i <= NF; i++) {
        printf "Field %d: %s\n", i, $i
    }
}'
```

Output

```
yaml

Field 1: Name
Field 2: Age
Field 3: Gender
Field 4: Country
```

Or if you just want the header line as-is:

```
bash
```

```
awk 'NR==1 { print }'
```

Or to store it in a variable for later use:

```
bash

awk 'NR==1 {
    header = $0
    print "Header is:", header
}'
```

You can also combine it with field labels like this:

```
bash

awk 'NR==1 {
    for (i = 1; i <= NF; i++) headers[i] = $i
    print "Header:"
    for (i = 1; i <= NF; i++) print headers[i]
}'
```

Want to use this header info later in the script too?

```
BEGIN{
    FS=","
    OFS="\t"
}
NR==1{
    for(i=1; i<=NF; i++) {
        # if(i==NF) printf $i;
        # else printf "%s\t",$i
        printf "%s\t", $i;
    }
    print "Average"
}
NR>1{
}
```

You're off to a good start! Your `awk` script is already correctly:

- Setting `FS` to `,` (for CSV input)
- Setting `OFS` to `\t` (for tab-separated output)

- Dynamically printing headers with an additional `"Average"` column

Let me help you clean up and complete it a little:

Full Working Version:

```
awk

BEGIN {
    FS = ","          # Input field separator is comma
    OFS = "\t"        # Output field separator is tab
}

NR == 1 {
    for (i = 1; i <= NF; i++) {
        printf "%s\t", $i
    }
    print "Average"
}

NR > 1 {
    sum = 0
    count = 0
    for (i = 2; i <= NF; i++) { # assuming 1st column is Name or ID
        sum += $i
        count++
        printf "%s\t", $i
    }
    avg = sum / count
    printf "%.2f\n", avg
}
```

Sample Input:

```
csv

Name,Math,Science,English
Alice,80,90,85
Bob,70,75,65
```

Output:

javascript

Name	Math	Science	English	Average
Alice	80	90	85	85.00
Bob	70	75	65	70.00

This script calculates the **average of all numeric fields** except the first one (assuming it's a name), and prints it as the last column.

Want to customize which columns to include in the average?

this is my input.csv:

```
Name,MA110,CS104,MS101,BB101
Kavya,98,87,99,91
Saksham,100,80,90,95
Priya,90,75,81,96
```

but the command is printing this:

```
Name Average CS104 MS101 BB101
```

Ah, I see the issue. The problem lies in the way the fields are being printed in the `NR > 1` block. In your current script, you're **not printing the first field (Name)**, and you're starting your loop at `i = 2`, which skips over it.

Fix

In `NR > 1`, you should:

1. Print the **name field** (`$1`) first.
2. Then loop through **all numeric fields, starting from field 2** to the end, to:
 - Print the marks
 - Calculate the sum
3. Finally, print the average.

✓ Corrected AWK Script

awk

```

BEGIN {
    FS = ","
    OFS = "\t"
}

NR == 1 {
    for (i = 1; i <= NF; i++) {
        printf "%s\t", $i
    }
    print "Average"
}

NR > 1 {
    printf "%s\t", $1    # Print Name
    sum = 0
    count = 0
    for (i = 2; i <= NF; i++) {
        printf "%s\t", $i
        sum += $i
        count++
    }
    avg = sum / count
    printf "%.2f\n", avg
}

```

Output for your input:

tsv

Name	MA110	CS104	MS101	BB101	Average
Kavya	98	87	99	91	93.75
Saksham	100	80	90	95	91.25
Priya	90	75	81	96	85.50

Let me know if you want to sort the output by average or write it to a new file!

how to remove the carriage return of the csv file

To remove **carriage return characters** (`\r`) from a CSV file—common if the file was created on Windows—you can use several simple command-line tools:

✓ 1. Using `dos2unix` (best option if available)

```
bash

dos2unix input.csv
```

This modifies the file in-place, converting Windows line endings (`\r\n`) to Unix (`\n`).

✓ 2. Using `tr` (translate/delete)

```
bash

tr -d '\r' < input.csv > output.csv
```

This removes all `\r` characters and writes to `output.csv`.

✓ 3. Using `sed`

```
bash

sed 's/\r//' input.csv > output.csv
```

Or in-place (GNU `sed` only):

```
bash

sed -i 's/\r//' input.csv
```

✓ 4. Using `awk` (if you prefer)

```
bash
```

```
awk '{ sub(/\r$/, ""); print }' input.csv > output.csv
```

Check for carriage returns:

If you're not sure whether `\r` characters are present:

```
bash
```

```
cat -v input.csv
```

Lines ending with `^M` indicate carriage returns.

Want to remove them in-place *only if present*?